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- Spring 26
- BAN 650
- Assignment 7
- Assignment Open Date: March 25, 2026
- Assignment Due Date: April 1, 2026

Matrix (Stock Level by Store Format & Department)

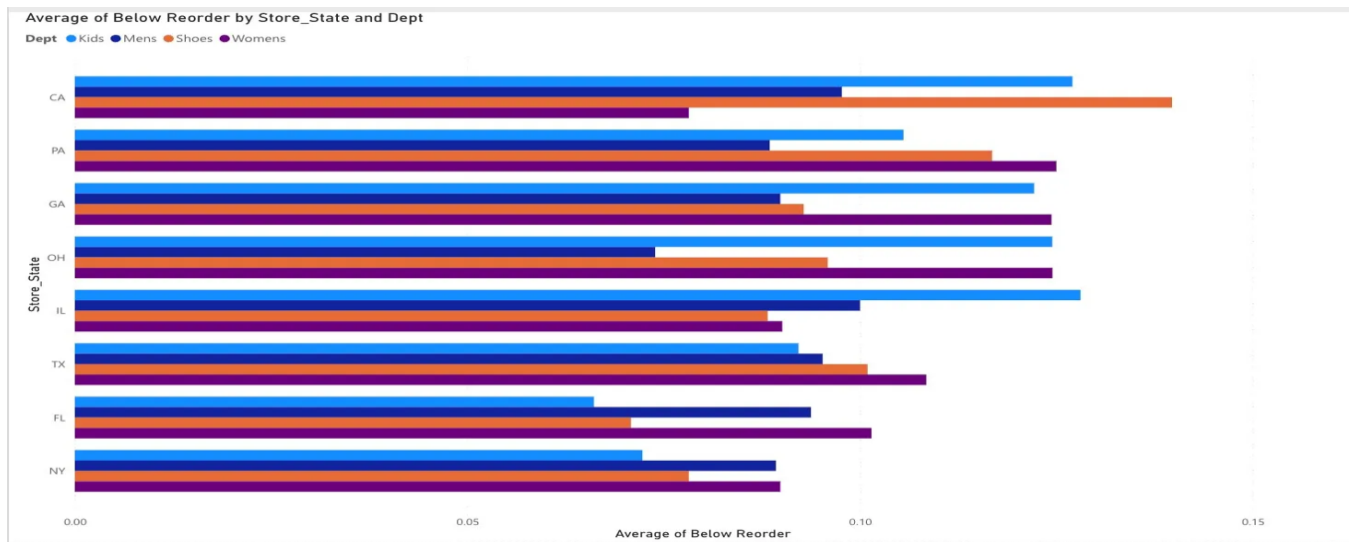
Store_Format	Kids	Mens	Shoes	Womens	Total
Express	248.62	239.65	249.22	255.39	248.26
Mall	246.03	254.04	246.33	254.40	250.22
Supercenter	241.61	259.56	254.36	242.01	249.09
Total	245.45	251.08	249.85	250.29	249.19

What relationships or dependencies are being shown: The matrix shows the relationship between store format (Express, Mall, Supercenter) and department (Kids, Mens, Shoes, Womens) in terms of average stock levels. The color gradient immediately surfaces which format-department combinations are well-stocked (green) versus at risk (red).

Why these relationships matter from a managerial perspective: A store format that consistently underperforms in a specific department signals a structural inventory gap. For example, Supercenter-Womens (242.01) and Express-Mens (239.65) are the reddest cells, meaning those combinations are most vulnerable to stockouts, which directly impacts sales and customer satisfaction.

What potential action or question the visualization raises: Should reorder thresholds be recalibrated differently per store format? A Supercenter likely has higher foot traffic and demand than an Express store, so a stock level of 242 may be far more critical there than the same number at a smaller format.

Task 2 Bar Chart (Stockout Risk by State & Department)



What relationships or dependencies are being shown: The chart shows the dependency between geographic location and department-level stockout risk, measured as the proportion of SKUs falling below their reorder point. CA and PA lead across nearly all departments, with CA-Shoes (0.14) being the single highest risk combination.

Why these relationships matter from a managerial perspective: When multiple departments in the same state all show elevated stockout risk simultaneously, it points to a regional supply chain problem rather than a department-specific issue. This makes it a systemic risk that affects the entire state's inventory health at once.

What potential action or question the visualization raises: Are CA and PA stores being served by the same distribution center or supplier? If so, that supplier or DC is likely the bottleneck — and addressing it would improve stockout rates across all four departments in those states at once.